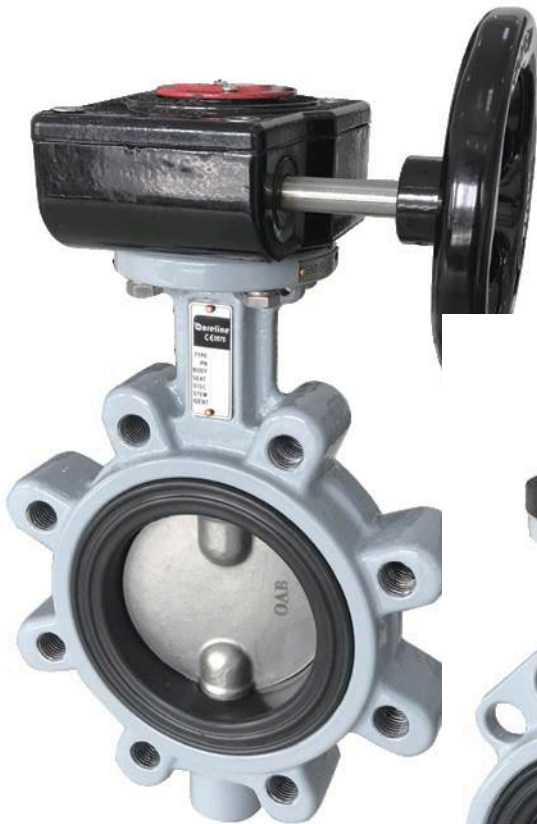


Rubber seat butterfly valves

Fig.223 : Wafer

Fig.224 : Lug



General applications and specifications

General applications

Food and beverage industries
Pneumatic materials handling technologies
Shipbuilding
Water treatments
Power plants
Chemical and petrochemical industries
Pharmaceutical industries
Painting industries
Civil engineering
Pulp and paper industries



Specifications

Nominal diameter:	DN40-DN2000
Standard differential pressure:	16 bar for DN40-DN200, 10 bar for DN250-DN2000
Max. differential pressure:	20 bar for DN40-DN600 16 bar for DN700-DN2000
Vacuum:	0.001 bar absolute (depending on medium and temperature)
Flange accommodation:	DN40-DN300: EN1092 PN10, PN16. ASME B16.5 Class150. JIS B 2239 10K, 16K. BS10 Table D, Table E DN350-DN2000: EN1092 PN10, PN16. ASME B16.5 Class150. BS10 Table D, Table E
Face to face:	EN558 Series 20; API 609 Table 1
Top flange:	EN ISO 5211
Temperature range:	-20°C to +180°C (depending on pressure, medium and material)
Tightness test:	ISO 5208 Rate A, API 598 Table 5 (medium: water)



The butterfly valves meet the safety requirements of the pressure Equipments Directive 97/23/EC (PED) appendix 1 for fluids of the groups 1 and 2.



Design features and chart for medium resistance

Design features DN40 to DN300



Groove to show direction of the disc. Heat number and material certificate. Threaded to make safe mounting of hand lever.

Top flange according to ISO5211, together with square stem, makes it suitable for direct mounting of actuators, without using bracket.

Relief track on the mounting flange protect the actuator if over pressure or wear makes the valve leak.

Stainless steel retainer preventing stem blow-out.

Weather sealing.

Unique detail make it convenient for adding identification or manual book.

RPTFE/graphite bearings provide excellent support.

GGG40 Ductile iron body, provides high mechanical properties.

O-rings by upper and lower shaft, are extra ordinary sealing to prevent leakage.

Square transmission between disc and shaft is strong and reliable.

UPR hard back ring ensure good tolerances of sealing point, keep the rubber stationary even under vacuum conditions and let you mount the valve between flanges without seat displacement.

Streamline disc for better Kv with enough strength.

Unique curve sloop reduce the torque and increase life time.

Wider diameter of sealing, make it safe for all kind of flanges.

Patented solution on disc turning point with reinforced bearing inside the disc, where rotation occurs. This reduce friction and ensure perfect centering and long lifetime together with the strong fixing shaft.

Multi-standard alignment holes suitable for various standard of flanges: EN1092 PN10,PN16, ASME B16.5 Class150, JIS B2239 10K,16K, BS10 Table D, Table E.

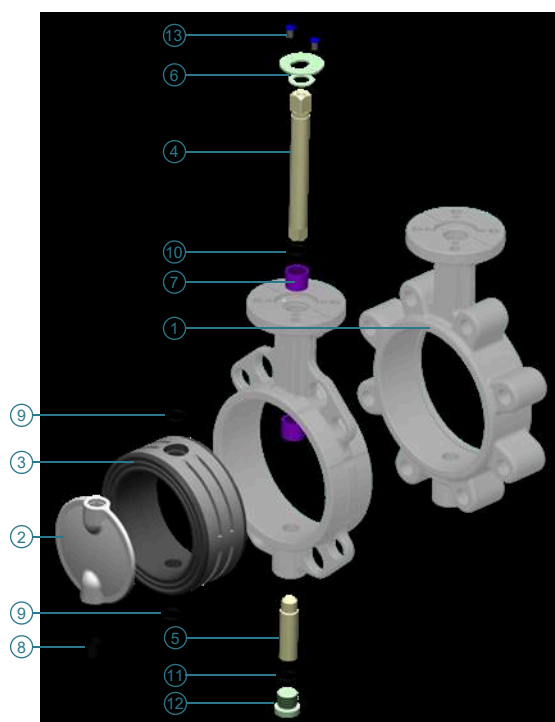
Plug with O-ring as wheather and secondary sealing.

Chart for medium resistance

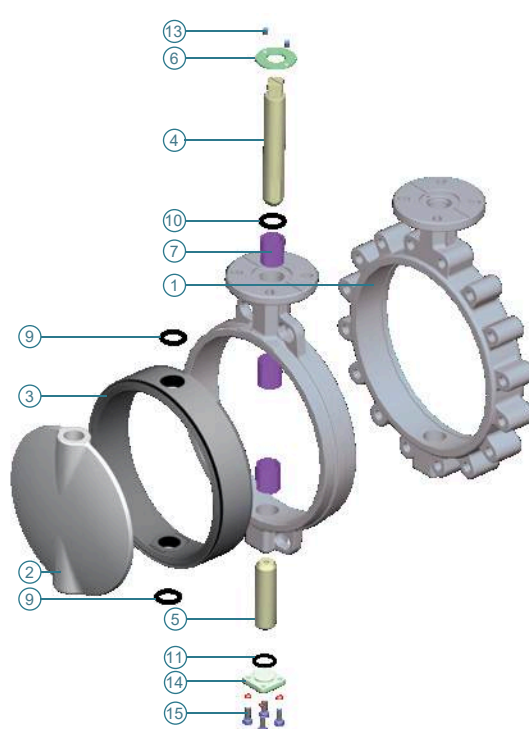
Liner	Suitable for	Unsuitable for
EPDM	Water, steam, alcohol, glycol, caustic soda, ozone, food products, glycerine, milk, oxygen, air, saturated salt, iron chloride, gelatine, dry hydrogen sulphide, potassium chloride, sodium, magnesium chloride	Mineral oil, chlorine compounds, ketones, acetyl, chloride, asphalt, bromine, butane, butyl, petrol, diesel oil, acid, fish oil, freon, chlorine, natural gas, exhaust gas, nitric acid
NBR	Mineral oil, grease, air, seawater, gas, boric acid, aluminium chloride, ammonia gas, citric acid, diesel oil, fish oil, petrol, gelatine, glycerine, magnesium chloride, lactic acid, linseed oil, natural gas	Ozone, acetone, aniline, chlorine dioxide, chromic acid, phenol, ethyl acetate, freon 21+22+23, hot nitric acid, styrene, hydrogen sulphide, isopropyl acetate, oxygen, sulphuric acid
NATURAL RUBBER	Abrasive medium, aluminium chloride, sugar beet fluid, boric acid, potassium chloride, citric acid, magnesium chloride, ferric nitrate, formic acid, gelatine, sugar, glycerine, lactic acid, nitrogen	Steam, aniline, asphalt, butadiene, diesel oil, ethane, ethyl acetate, hydraulic oil, hydrochloric acid, linseed oil, methane, mineral oil, oxygen, styrene, soyabean oil, turpentine
HYPALON	Mineral, organic and inorganic chemicals, air, oxygen, fish oil, glycerine, citric acid, ozone, sodium sulphate	Ammonia, diesel oil, grease, ketone, methyl, phenol, propyl, bromine, nitric acid, tar, urea, varnish, lectine
SILICONE	Food products, ammonia gas, barium, boric acid, potassium, bisulphite, citric acid, copper cyanide, glycerine, nitrogen, lactic acid	Steam and hot water (max. 100° C), asphalt, diesel oil, ethane, freon, ethyl chloride, methane, nitric acid, olive oil, propane, turpentine
FPM	Oil, mineral acid, grease, phosphorus, tannic acid, gelatine, glycol, oxygen, slaked lime, carbon acid, natural gas, pulp, salt, sugar, sulphur	Hot water, steam, ketone, ammonia gas, acetone, formaldehyde, cellulose acetate, freon, urea, ethanoic acid, methyl
FPM-B	Acid, alkali, amine hot water, steam	Gasoline, naphtha, hydrocarbon solvent, chlorine solvent
PTFE	Almost available for all medium applications	None

Parts list and material specification

DN40-DN300



DN350-DN2000



No.	Part name	Material	Specification	No.	Part name	Material
1	Body	Ductile iron	EN1563 JS1030	4/5	Stem	SS420
2	Disc	Stainless steel	ASTM A351 CF8	4/5	Stem	SS431
			ASTM A351 CF8M			17-4PH SS
			ASTM A351 CF3M	6	Preventing plate	SS304
			2507			SS316
		Alloy steel	1.4462	7	Body bearing	RPTFE with graphite
			1.4529	8	Disc bearing	Rainforced nylon
			1.4539	9	O ring	Same as seat
			Hastelloy	10	Weather seal	NBR
3	Seat *)	Aluminium bronze	C95800	11	Anti-dust seal	NBR
		Ductile iron	Rilsan, Halar coated	12	Plug	Steel galvanized
		NBR (Eq. Nitrile)	-15°C~+85°C	13	Screw	SS304
		X-NBR ¹⁾	-15°C~+85°C	13	Screw	SS316
		NBR-DVGW ²⁾	-15°C~+60°C			
		EPDM-H	-20°C~+125°C	14	Bottom cover	Same as body
		EPDM-FDA (white)	-20°C~+85°C	15	Bolt	SS304
		EPDM-FDA (black)	-20°C~+125°C			SS316
		FPM (Eq. FKM, Viton)	-15°C~+150°C			
		FPM-B (Eq. FEPM) ³⁾	-15°C~+150°C			
		PTFE	-15°C~+150°C ⁴⁾			

Notes:

*) Rubber seat with hard UPR backup.

1) Well -resistant NBR.

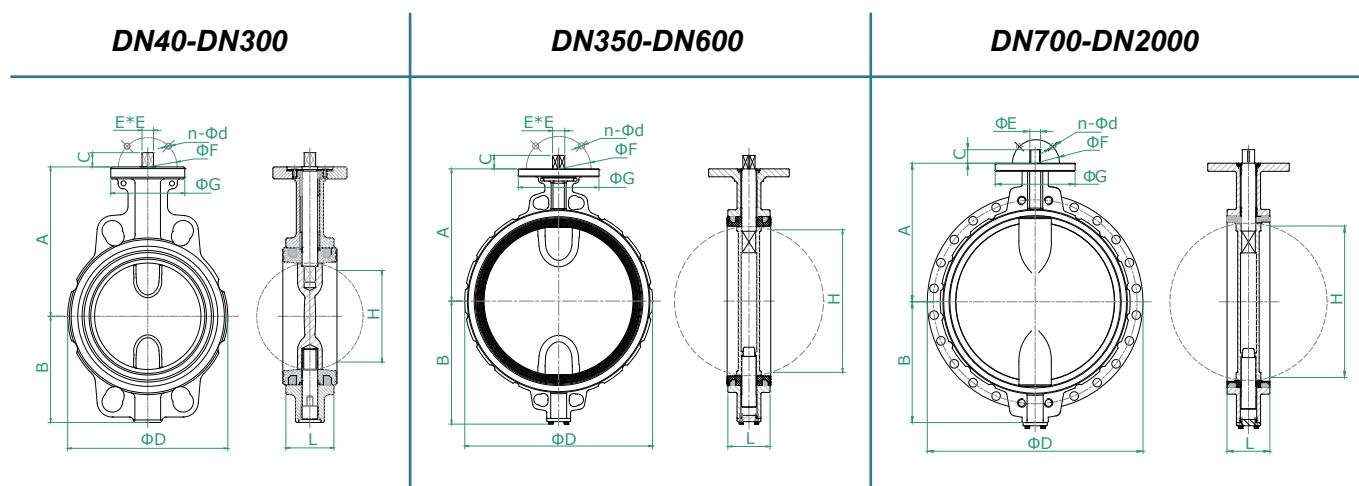
2) German gas certificate.

3) Steam resistant FPM.

4) Depending on the backup rubber material, available with EPDM and FPM backup.

The above temperature range for the valve seats are provided as reference for general working conditions. Please note that the actual applications may vary due to the different media, pressure etc. in the pipeline.

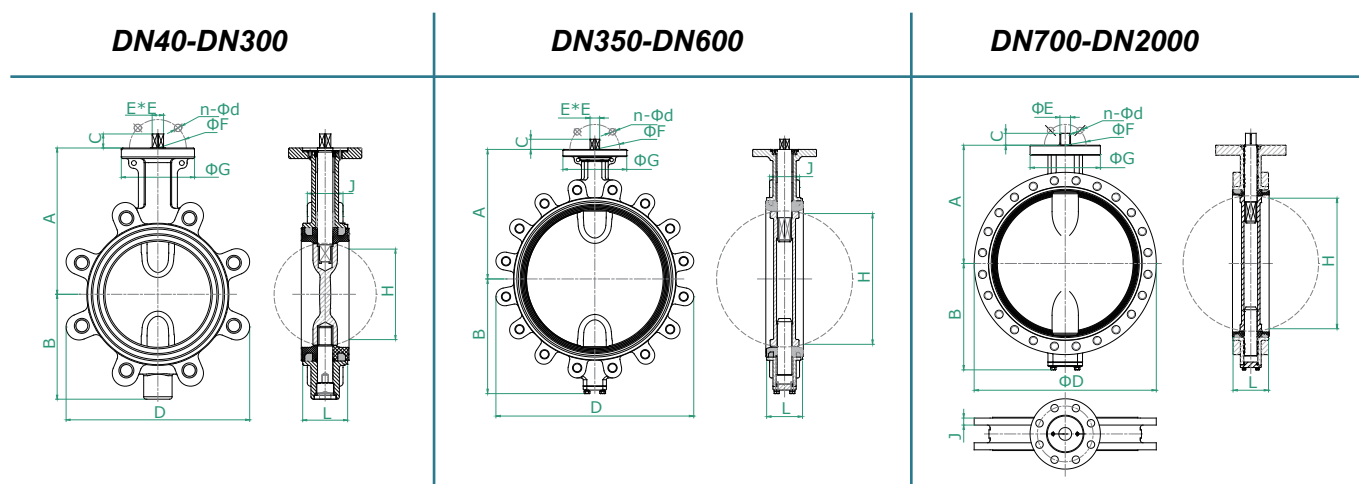
Fig. 223 Dimensions



Size	A	B	C	D	E	F	n	d	G	H	L	[kg]
DN25/32	108	60	13.5	72	11	50	4	7	65	0	32	1.4
DN40	113	67.5	13.5	86	11	50	4	7	65	30	33	1.7
DN50	126	76	13.5	102	11	50	4	7	65	35	43	2
DN65	134	82	13.5	116	11	50	4	7	65	47	46	2.6
DN80	157	95.5	13.5	132	11	50	4	7	65	70	46	3.3
DN100	167	113.5	17.5	157	14	50+70	4	7+9	90	87	52	5
DN125	180	129	17.5	195	14	70	4	9	90	117	56	6.4
DN150	203	142	18.5	218	17	70	4	9	90	144	56	7.8
DN200	228	172	24.5	271	22	102	4	11	125	191	60	12.2
DN250	266	213	24.5	329	22	102	4	11	125	241	68	19
DN300	291	242	26.5	382	27	102+125	4+4	12+14	150	291	78	26
DN350	332	273	30	422	27	125+140	4+4	14+18	175	329	78	41
DN400	363	317	30	484	27	125+140	4+4	14+18	175	376	102	58
DN450	397	348	39	542	36	140+165	4+4	18+22	210	425	114	80
DN500	425	393	49	597	46	140+165	4+4	18+22	210	475	127	97
DN600	498	453	49	708	46	165+254	4+8	22+18	300	573	154	169
DN700	626	531	90	928	63.1	254	8	18	300	674	165	252
DN750	660	564	90	984	63.1	254	8	18	300	727	165	290
DN800	666	601	90	1061	63.1	254	8	18	300	771	190	367
DN900	722	660	110	1170	74.7	254	8	18	300	839	203	465
DN1000	806	728	120	1290	83.7	298	8	22	350	939	216	606
DN1100	826	771	140	1404	94.7	298	8	22	350	1036	255	805
DN1200	941	874	150	1511	104.7	298	8	22	350	1137	276	900
DN1400	1000	940	175	1685	139.9	356	8	32	415	1351	279	1158
DN1600	1155	1085	195	1930	160	356	8	32	415	1548	318	1684
DN1800	1200	1170	195	2170	174.5	406	8	39	475	1703	356	2645
DN2000	1363	1360	245	2345	199	406	8	39	475	1938	406	4000

Different pressure may cause different dimension of "D".

Fig. 224 Dimensions



DN	A	B	C	D	E	F	n	d	G	H	J	L	[kg]
40	113	67.5	13.5	113	11	50	4	7	65	30	30	33	2.5
50	126	76	13.5	117	11	50	4	7	65	35	34	43	2.8
65	134	82	13.5	131	11	50	4	7	65	47	36	46	3.7
80	157	95.5	13.5	176	11	50	4	7	65	70	36	46	5.1
100	167	113.5	17.5	206	14	50+70	4	7+9	90	87	40	52	6.9
125	180	129	17.5	236	14	70	4	9	90	117	44	56	9.3
150	203	142	18.5	258	17	70	4	9	90	144	44	56	10.5
200	228	172	24.5	321	22	102	4	11	125	191	50	60	17.4
250	266	213	24.5	395	22	102	4	11	125	241	57	68	28.1
300	291	242	26.5	461	27	102+125	4+4	12+14	150	291	66	78	40
350	332	273	30	511	27	125+140	4+4	14+18	175	329	62	78	55
400	363	317	30	580	27	125+140	4+4	14+18	175	376	84	102	85
450	397	348	39	630	36	140+165	4+4	18+22	210	425	94	114	114
500	425	393	49	700	46	140+165	4+4	18+22	210	475	110	127	144
600	498	453	49	823	46	165+254	4+8	22+18	300	573	120	154	227
700	626	531	90	928	63.1	254	8	18	300	674	54	165	342
750	660	564	90	984	63.1	254	8	18	300	727	54	165	400
800	666	601	90	1061	63.1	254	8	18	300	771	60	190	485
900	722	660	110	1170	74.7	254	8	18	300	839	60	203	605
1000	806	728	120	1290	83.7	298	8	22	350	939	60	216	776
1100	826	771	140	1404	94.7	298	8	22	350	1036	50	255	985
1200	941	874	150	1511	104.7	298	8	22	350	1137	62	276	1190
1400	1000	940	175	1685	139.9	356	8	32	415	1351	62	279	1380
1600	1155	1085	195	1930	160	356	8	32	415	1548	65	318	2054
1800	1200	1170	195	2170	174.5	406	8	39	475	1703	70	356	3075
2000	1363	1360	245	2345	199	406	8	39	475	1938	75	406	4500

Valve torques (N*M)

Size	Standard disc differential pressure						Increased PN16 disc	Increased PN20 disc	Reduced PN6 disc
	EPDM/ NBR/ FPM seat				FDA seat	PTFE seat	EPDM/ NBR/ FPM seat		
	ΔP=3bar	ΔP=6bar	ΔP=10bar	ΔP=16bar	ΔP=16bar	ΔP=10bar	ΔP=16bar	ΔP=20bar	ΔP=6bar
DN40	9	9	10	11	15			15	7
DN50	10	10	11	12	20	13		16	8
DN65	15	16	17	18	30	20		25	10
DN80	23	24	25	26	40	30		40	15
DN100	35	36	37	38	60	45		60	22
DN125	45	48	52	57	90	58		80	35
DN150	90	95	100	105	150	116		150	55
DN200	185	190	195	200	300	240		256	91
DN250	260	270	280		450	335	380	450	170
DN300	300	320	340		550	390	400	510	230
DN350	500	550	600				720	870	400
DN400	620	700	800				870	1100	500
DN450	920	1000	1200				1600	2000	700
DN500	1600	1900	2200				3700	5700	950
DN600	2200	2500	2800				4900	7800	1600
DN700	3300	3600	3900				7300		2520
DN750	4400	4800	5300				8900		3400
DN800	6200	6700	7300				11000		4700
DN900	7100	7700	8300				13000		5400
DN1000	10000	11000	12000				24000		7700
DN1100	14000	15000	16000				32000		11000
DN1200	17000	19000	21000				42000		14000
DN1400	21000	23000	25000				50000		17000
DN1600	30000	32000	34000				70000		23000
DN1800	33000	36000	38000				80000		26000
DN2000	36000	38000	40000				90000		30000

* The torque above are not including safety factor. Add a safety factor 1.3 for sizing or consult the factory.

Service and medium factor - Actuator Sizing

Service factor [SF]	Multiply by	Medium factor [MF]	Multiply by	Medium factor [MF]	Multiply by
ON/OFF operation	1.15	Lubricating liquid/gas	0.90	For dry service (Dry gas/air)	1.25
Modulating operation	1.25	Viscous Liquids, Molasses	1.30	Dirty air slurry, natural gas, dirty slurry,	1.50-1.80
*) 2 cycle/day "NC"	1.15	Degreasing liquid	1.25	Lime water, chlorin gas,oxygen, powder	1.50-1.80
**) 1 cycle/week "NC"	1.50	Saturated steam	1.20	Hydrodynamic torque	NA

OBS: Butterfly valve torque is 100% by 0° to 6° angle and 33% from 7° to 90° angle.

* Valve is completely closed and opened 2 times a day minimum.

** Valve is completely closed and opened only one time per week or longer.

Having a long period without maneuvering the valve, will increase the breakaway torque.

EXAMPLE OF ACTUATOR SIZING: Simple ON/OFF operation, Medium: Molasses,

Valve: 223 DN100. 1.15[SF] x 1.30[MF] x 38[Nm] = 56.8Nm (Sizing torque actuator)

Only choose one Service factor [SF] and one Medium factor [MF] when calculating the sizing torque.

Flow capacities and valve sizing

Kv values-valve sizing Coefficients (M³/H at 1Bar ΔP)

DN	10°	20°	30°	40°	50°	60°	70°	80°	90°
40		1	3	7	14	26	38	47	52
50		2	6	13	29	47	70	97	105
65		3	11	26	49	77	121	170	212
80		6	28	54	91	140	213	301	390
100		19	38	79	140	232	412	590	676
125		27	84	156	248	385	624	945	1120
150	7	51	129	224	363	572	977	1490	1798
200	22	114	229	401	639	1018	1755	2680	3100
250	33	171	334	634	970	1530	2650	4105	5200
300	49	250	490	925	1416	2231	3865	6351	7350
350	118	301	631	1131	1918	3081	4963	8035	9993
400	153	393	824	1478	2506	4024	6482	10983	12595
450	195	498	1043	1871	3170	5093	8210	13695	16850
500	240	615	1288	2309	3913	6287	10128	17250	19306
600	345	885	1853	2958	5635	9054	14584	24980	28323
700	390	930	2210	3750	6959	11100	19200	33080	39700
750	450	1160	2400	4350	7890	12900	21200	36750	45350
800	520	1330	2650	5030	8890	14350	23750	39900	49530
900	600	1680	3350	6470	11890	19520	31700	52750	62000
1000	710	2210	4300	8100	15130	23720	40050	67000	81000
1100	990	3020	5980	10050	17580	28970	48950	83500	10580
1200	1278	4050	7650	12600	20100	34500	56750	99570	121000
1400	1470	5180	10100	18150	32000	51370	88900	142100	161000
1600	1730	7200	14530	26530	45400	71500	118700	198000	229000
1800	2020	8100	18750	34230	52130	90350	143780	243250	280000
2000	2320	9000	21320	40350	57980	110200	169870	278980	320000

Valve Sizing

Determining the size of butterfly valves for control purposes should not be done on the basis of the nominal diameter of the pipe but should be calculated on the basis of the operating characteristics in order to attain the correct control characteristics.

Butterfly valves Fig. 223 / Fig. 224 are designed with approximately equal percentage characteristics over an opening angle of 65°.

You only need to consider the opening angle when determining the size of control valves. When determining the valve nominal diameter calculate the Kv value from the below formula:

Liquid:

$$K_v = Q \times \sqrt{\frac{W}{\Delta p}}$$

Gas:

$$K_v = \frac{V_N}{514} \sqrt{\frac{G \times T}{\Delta p \times p_d}}$$

Kv = Flow coefficient

Q = Max. flow volume in m³/h

w = Exact weight in kg/m³

Δp = Pressure drop in bar

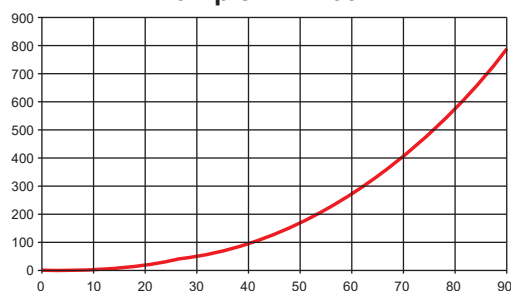
V_N = Max. flow in Nm³/h

G = Exact weight in kg/Nm³

T = Absolute temp. in ° Kelvin

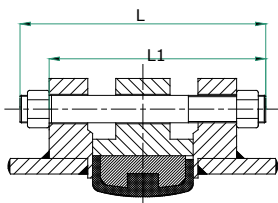
P_d = Absolute pressure downstream in bar

Example: DN 100

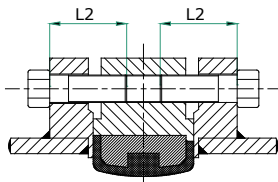


Bolt quantity and length for valve installation

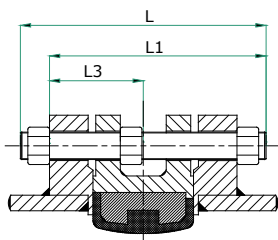
223 DN40-DN2000



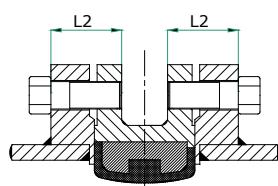
224 DN40-DN600



224 DN700-DN2000 Type 1 Connection

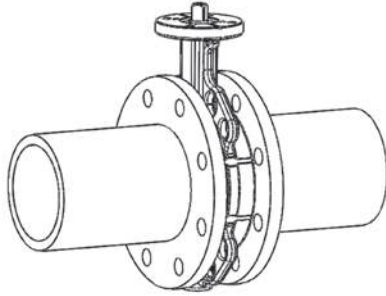


224 DN700-DN2000 Type 2 Connection



DN	Flange	Wafer type connection				Lug type connection		
		Bolt Dia.	Qty of bolts	L	L1	Qty of bolts	L2	L3
40	PN10	M16	4	135	115	4x2	35	
	ANSI150	1/2" - 10 UNC	4	120	100	4x2	35	
50	PN10	M16	4	145	125	4x2	40	
	ANSI150	5/8" - 11 UNC	4	130	110	4x2	40	
65	PN10	M16	4	150	130	4x2	40	
	ANSI150	5/8" - 11 UNC	4	140	120	4x2	45	
80	PN10	M16	8	150	130	8x2	40	
	ANSI150	5/8" - 11 UNC	4	145	125	4x2	45	
100	PN10	M16	8	160	140	8x2	45	
	ANSI150	5/8" - 11 UNC	8	150	130	8x2	45	
125	PN10	M16	8	165	145	8x2	50	
	ANSI150	3/4" - 10 UNC	8	155	130	8x2	50	
150	PN10	M20	8	175	150	8x2	50	
	ANSI150	3/4" - 10 UNC	8	160	135	8x2	50	
200	PN10	M20	8	180	155	8x2	55	
	PN16	M20	12	185	160	12x2	55	
	ANSI150	3/4" - 10 UNC	8	170	145	8x2	55	
	PN10	M20	12	200	175	12x2	60	
250	PN16	M24	12	210	185	12x2	60	
	ANSI150	7/8" - 9 UNC	12	190	165	12x2	60	
300	PN10	M20	12	210	185	12x2	65	
	PN16	M24	12	230	200	12x2	65	
	ANSI150	7/8" - 9 UNC	12	200	175	12x2	60	
	PN10	M20	16	215	190	16x2	65	
350	PN16	M24	16	240	210	16x2	65	
	ANSI150	1" - 8 UNC	12	220	190	12x2	65	
400	PN10	M24	16	255	230	16x2	75	
	PN16	M27	16	280	250	16x2	75	
	ANSI150	1" - 8 UNC	16	245	215	16x2	80	
	PN10	M24	20	275	245	20x2	80	
450	PN16	M27	20	310	280	20x2	80	
	ANSI150	1 1/8"-7UNC	16	265	235	16x2	80	
500	PN10	M24	20	295	260	20x2	85	
	PN16	M30	20	335	300	20x2	90	
	ANSI150	1 1/8"-7UNC	20	285	250	20x2	90	
	PN10	M27	20	310	280	20x2	100	
600	PN16	M33	20	385	350	20x2	100	
	ANSI150	1 1/4"-7UNC	20	335	300	20x2	110	
700	PN10	M27	24	335	300	24x2	100	130
	PN16	M33	24	370	330	24x2	115	150
	ANSI150	1 1/4"-7UNC	28	385	345	28x2	120	160
	PN10	M30	24	375	340	24x2	115	150
800	PN16	M36	24	425	380	24x2	130	175
	ANSI150	1 1/2"-6UNC	28	440	395	28x2	135	185
900	PN10	M30	28	400	365	28x2	115	155
	PN16	M36	28	455	410	28x2	135	185
	ANSI150	1 1/2"-6UNC	32	475	430	32x2	145	195
	PN10	M33	28	435	395	28x2	125	165
1,000	PN16	M39	28	485	440	28x2	145	190
	ANSI150	1 1/2"-6 UNC	36	485	440	36x2	145	190
1,200	PN10	M36	32	505	365	32x2	145	195
	PN16	M45	32	505	460	32x2	145	195
	ANSI150	1 1/2"-6UNC	44	560	515	44x2	170	220
	PN10	M39	36	500	455	36x2	120	175
1,400	PN16	M45	36	545	495	36x2	140	195
	ANSI150	1 3/4"-5UNC	48	620	575	48x2	180	235
1,600	PN10	M45	40	565	515	40x2	135	190
	PN16	M52	40	630	575	40x2	160	220
	ANSI150	1 3/4"-5UNC	52	565	515	52x2	135	190
	PN10	M45	44	620	575	44x2	150	205
1,800	PN16	M52	44	685	630	44x2	175	230
	ANSI150	1 3/4"-5UNC	60	620	575	60x2	150	205
2,000	PN10	M45	48	680	635	48x2	160	215
	PN16	M56	48	770	710	48x2	190	250
	ANSI150	2"-4.5UNC	64	680	635	64x2	160	215

Installation Guide



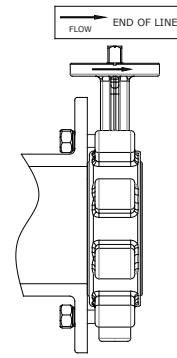
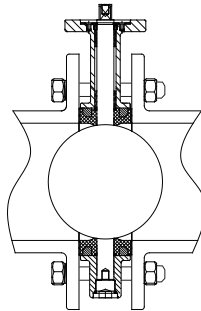
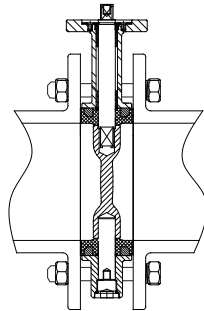
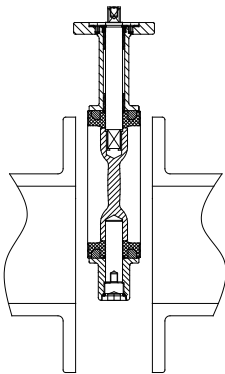
- ☐ Check that the specifications on the identification plate meet the requirements regarding pressure, temperature and media.
- ☐ The piping must have a straight line and the flanges have to be parallel. Furthermore there must be a distance between the flanges, corresponding to the face-to-face dimensions of the butterfly valve.
- ☐ The butterfly valve can be mounted in any direction. However if there are a lot of dirt particles on the bottom of the pipe, it will be suitable to mount the disc.
- ☐ Before commissioning, the pipework has to be rinsed out to remove dirt and remnants of welding material, to avoid damage on the liner. During the rinsing procedure, the butterfly valve has to be positioned as open and may not be operated before the rinsing has been completed.
- ☐ Welding operations may not be performed nearby the butterfly valve, as welding drops can damage the liner.
- ☐ Do not use gaskets. The liner works as sealing to the atmosphere.
- ☐ Where vacuum, high flow rate or water hammering can occur, flanges without a loose collar should be used, to obtain the best conditions.

Carefully place the butterfly valve between the flanges, with the disc in closed position.

Check that the flange covers the area of the liner. Afterwards tighten the bolt on the flange by hand.

Carefully open and close the valve to check that the disc centralizes and the disc does not touch the flange. With the disc in the open position, tighten crosswise with a wrench.

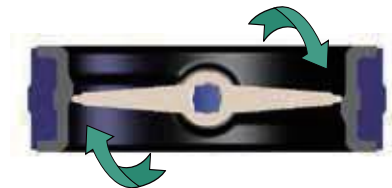
Ensure that 2240 lug butterfly valve is installed in proper flow direction for end of pipe service: the flow arrow direction on valve nameplate or body should be in accordance with the flow direction of medium.



As the butterfly valves are equipped with the unique wave shaped liner, the operation of the valves, either free stem, handle or gear operated, has to follow the guidelines as shown below.



A small triangular shaped figure is placed on the liner, this triangle indicates which way the disc opening. Turn disc Anticlockwise towards arrow to open



Turn Clockwise to close the valve, the valve sealing in the area of -2° to 2° range. If lower torque necessary, stop the disc to the 2° to 0° range; If tighter sealing necessary, stop the disc to the 0° to -2° range.

The content or parameters are indicative and can be changed without any notice.